

# Le système international d'unités The International System of Units

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Annexe 2

## Annex 2

Valeurs recommandées des fréquences étalons  
destinées à la mise en pratique de la définition du  
mètre et aux représentations secondaires de la  
seconde

**Recommended values of standard frequencies  
for applications including the practical  
realization of the metre and secondary  
representations of the second**

Partie 2

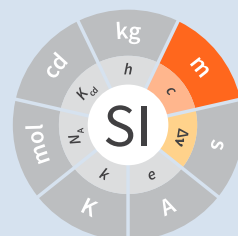
## Part 2

Ytterbium 171 Atom ()

**Ytterbium 171 Atom ()**

CCL-CCTF Frequency Standards Working Group

August 2018  
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**Recommended values of standard  
frequencies for applications including  
the practical realization of the metre and  
secondary representations of the second  
Part 2: Ytterbium 171 Atom ()**

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## 1. Recommended value [1] of the frequency in the CIPM List of Frequencies

$$f(^{171}\text{Yb}) = 518\,295\,836\,590\,863.6\text{ Hz}$$

equivalent to

$$\lambda(^{171}\text{Yb}) = 578\,419\,575.916\,163\text{ fm},$$

with a relative standard uncertainty of  $5 \times 10^{-16}$ .

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second [2].

## 2. Source data

The adopted value of  $f(^{171}\text{Yb}) = 518\,295\,836\,590\,863.6$  Hz with a relative uncertainty of  $5 \times 10^{-16}$  has been calculated by the methods presented in [3], [4], [5], [6], with the data from the table of evaluated frequencies (2017) attached below. The estimated relative standard uncertainty of  $5 \times 10^{-16}$  takes into account the relative uncertainty of the  $^{87}\text{Sr}$  value of  $4 \times 10^{-16}$  governing the evaluation procedure and also takes into account the smaller database of the  $^{171}\text{Yb}$  measurements.

**Table 1.** Table of evaluated frequency and frequency ratio measurements (2017)

| Frequency / Hz           | Uncertainty / Hz | Reference  |
|--------------------------|------------------|------------|
| $^{199}\text{Hg}$        |                  |            |
| 1 128 575 290 808 155.1  | 6.4              | [7]        |
| 1 128 575 290 808 154.62 | 0.41             | [8]        |
| $^{171}\text{Yb}$        |                  |            |
| 518 295 836 590 864.0    | 28.0             | [38]       |
| 518 295 836 590 863.1    | 2.0              | [39]       |
| 518 295 836 590 865.2    | 0.7              | [40]       |
| 518 295 836 590 863.5    | 8.1              | [41]       |
| 518 295 836 590 863.59   | 0.31             | [42]       |
| 518 295 836 590 863.38   | 0.57             | [43]       |
| $^{88}\text{Sr}$         |                  |            |
| 429 228 066 418 009.0    | 32.0             | [9]        |
| 429 228 066 418 008.3    | 2.1              | [10], Sr1  |
| 429 228 066 418 007.3    | 2.9              | [10], Sr2  |
| $^{87}\text{Sr}$         |                  |            |
| 429 228 004 229 874.0    | 1.1              | [11]       |
| 429 228 004 229 873.65   | 0.37             | [12]       |
| 429 228 004 229 873.6    | 1.1              | [13]       |
| 429 228 004 229 874.1    | 2.4              | [14]       |
| 429 228 004 229 872.9    | 0.5              | [15]       |
| 429 228 004 229 873.9    | 1.4              | [16]       |
| 429 228 004 229 872.0    | 1.6              | [17]       |
| 429 228 004 229 873.56   | 0.49             | [18]       |
| 429 228 004 229 873.7    | 1.4              | [19]       |
| 429 228 004 229 873.13   | 0.17             | [20]       |
| 429 228 004 229 873.10   | 0.13             | [21]       |
| 429 228 004 229 872.92   | 0.12             | [22]       |
| 429 228 004 229 872.97   | 0.16             | [23], 2014 |

| Frequency / Hz                    | Uncertainty / Hz       | Reference       |
|-----------------------------------|------------------------|-----------------|
| 429 228 004 229 873.04            | 0.11                   | [23], June 2015 |
| 429 228 004 229 872.97            | 0.40                   | [24]            |
| 429 228 004 229 872.99            | 0.18                   | [25]            |
| $^{40}\text{Ca}^+$                |                        |                 |
| 411 042 129 776 393.2             | 3.0 <sup>(a)</sup>     | [26]            |
| 411 042 129 776 398.4             | 1.2                    | [27]            |
| 411 042 129 776 400.5             | 1.2                    | [28], 2012      |
| 411 042 129 776 401.7             | 1.1                    | [28], 2014/2015 |
| $^{87}\text{Rb}$                  |                        |                 |
| 6 834 682 610.904 312 9           | $2.4 \times 10^{-6}$   | [29]            |
| 6 834 682 610.904 307 0           | $3.1 \times 10^{-6}$   | [30]            |
| 6 834 682 610.904 312 5           | $2.1 \times 10^{-6}$   | [31]            |
| Frequency ratios                  | Fractional uncertainty | Reference       |
| $^{199}\text{Hg}/^{87}\text{Sr}$  |                        |                 |
| 2.629 314 209 898 909 60          | $8.4 \times 10^{-17}$  | [32]            |
| 2.629 314 209 898 909 15          | $1.8 \times 10^{-16}$  | [8]             |
| $^{199}\text{Hg}/^{87}\text{Rb}$  |                        |                 |
| 165 124.754 879 997 258           | $3.8 \times 10^{-16}$  | [8]             |
| $^{171}\text{Yb}/^{87}\text{Sr}$  |                        |                 |
| 1.207 507 039 343 341 2           | $1.4 \times 10^{-15}$  | [33], [34]      |
| 1.207 507 039 343 337 76          | $2.4 \times 10^{-16}$  | [35]            |
| 1.207 507 039 343 337 749         | $4.6 \times 10^{-17}$  | [36]            |
| $^{88}\text{Sr}/^{87}\text{Sr}$   |                        |                 |
| 1.000 000 144 883 682 777         | $7 \times 10^{-17(b)}$ | [37]            |
| $^{87}\text{Sr}/^{40}\text{Ca}^+$ |                        |                 |
| 1.044 243 334 529 641 6           | $2.3 \times 10^{-15}$  | [27]            |
| $^{87}\text{Sr}/^{87}\text{Rb}$   |                        |                 |
| 62 801.453 800 512 435            | $3.3 \times 10^{-16}$  | [22]            |

(a) From the least square procedure it turned out that the value and the uncertainty given in ref. [26] is not compatible with the remaining data. Thus the uncertainty given in the original publication [26] was increased to 3 Hz to make it statistically more consistent.

(b) The fractional uncertainty of  $2.3 \times 10^{-17}$  in ref. [37] has been increased by a factor of three since this uncertainty is an order of magnitude smaller than the other input data for this transition.

**$^{171}\text{Yb}$  neutral atom,  $6s^2\ ^1S_0 - 6s6p\ ^3P_0$  unperturbed optical transition**

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